



CHEMISTRY

9729/01

Paper 1 Multiple Choice

01 October 2020

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, civics group and registration number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

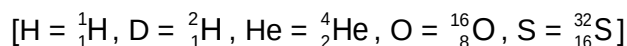
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this question paper.

The use of an approved scientific calculator is expected, where appropriate.

- 1 Which one of the following ions have more electrons than protons and more protons than neutrons?



- A S^{2-} B D_3O^+ C He^+ D OH^-

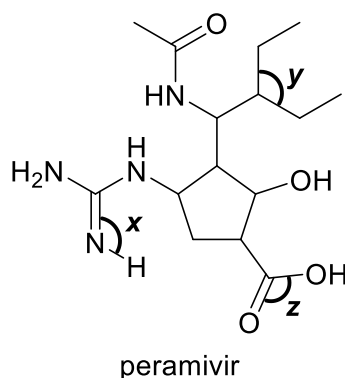
- 2 The first eight ionisation energies of an element are as follows (in kJ mol^{-1}):

703, 1610, 2460, 4350, 5400, 8500, 10300, 12300

The outer electronic configuration of the element is

- A ns^1 B ns^2 C $ns^2 np^2$ D $ns^2 np^3$

- 3 Peramivir is an anti-viral drug used for the treatment of patients infected by influenza viruses. It inhibits the action of a viral protein and prevents the release of viral particles from infected host cells. The structure of peramivir is given below.



What are the values of the bond angles x , y and z ?

	x	y	z
A	104.5°	109.5°	120°
B	107°	120°	119°
C	119°	109.5°	120°
D	120°	120°	119°

- 4 In which pair(s) of molecules does the **first** molecule have a higher boiling point than the **second** molecule?

- 1 pent-1-ene and 3-methylbut-1-ene
- 2 NO_2 and CO_2
- 3 NH_3 and HF

A 1, 2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 1 only

- 5 Which pair of molecules have the same shape?

- A** N_2O , SnCl_2
B AlCl_3 , BH_3
C SiF_4 , XeF_4
D BrF_5 , PCl_5

- 6 Which of the following contains one mole of the stated particles?

- A** Iodide ions in 0.5 mol solid silver iodide
B Nitrogen molecules in 14 g of nitrogen gas
C Carbon atoms in 8.0 dm^3 of propane gas at r.t.p
D Hydrogen ions in 1 dm^3 of 1 mol dm^{-3} aqueous sulfuric acid

- 7 10 cm^3 of a gaseous hydrocarbon was completely burnt in 55 cm^3 of oxygen. The volume of residual gas obtained was 40 cm^3 . When the residual gas was shaken with aqueous sodium hydroxide, the volume decreased to 10 cm^3 . All measurements were taken at room temperature and pressure.

Which of the following represents the hydrocarbon?

- A** C_3H_4 **B** C_3H_6 **C** C_3H_8 **D** C_3H_{10}

- 8 Sodium thiosulfate is used in the textile industry to remove an excess of chlorine from bleaching processes by reducing it to chloride ions.



In this reaction, how many moles of electrons are supplied per mole of thiosulfate?

- A 1 B 2 C 4 D 8
- 9 A sample of 0.20 g of liquid E is vapourised in a graduated syringe at 100 °C and 1 bar. The final volume of the syringe recorded was 38 cm³.

By assuming that 1 mol of vapour occupies 22.7 dm³ at standard temperature and pressure, what is the relative molecular mass of E?

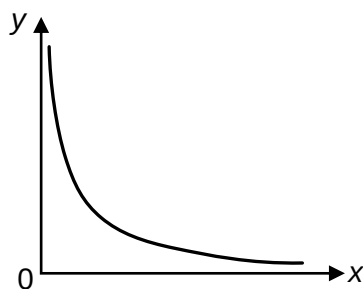
A $\frac{0.038 \times 273 \times 22.7}{0.20 \times 373}$

B $\frac{0.038 \times 373 \times 22.7}{0.20 \times 273}$

C $\frac{0.20 \times 273 \times 22.7}{0.038 \times 373}$

D $\frac{0.20 \times 373 \times 22.7}{0.038 \times 273}$

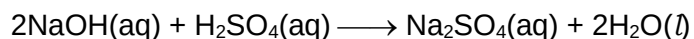
- 10 For a fixed mass of an ideal gas, which of the following could give the graph shown below?



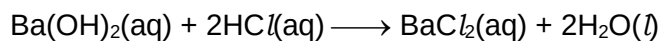
- 1 p against concentration (at constant T)
- 2 V against p (at constant T)
- 3 density against T (at constant p)

- A 1, 2 and 3 B 1 and 2 only C 2 and 3 only D 1 only

- 11 The heat liberated in the neutralisation given below is 114 kJ.

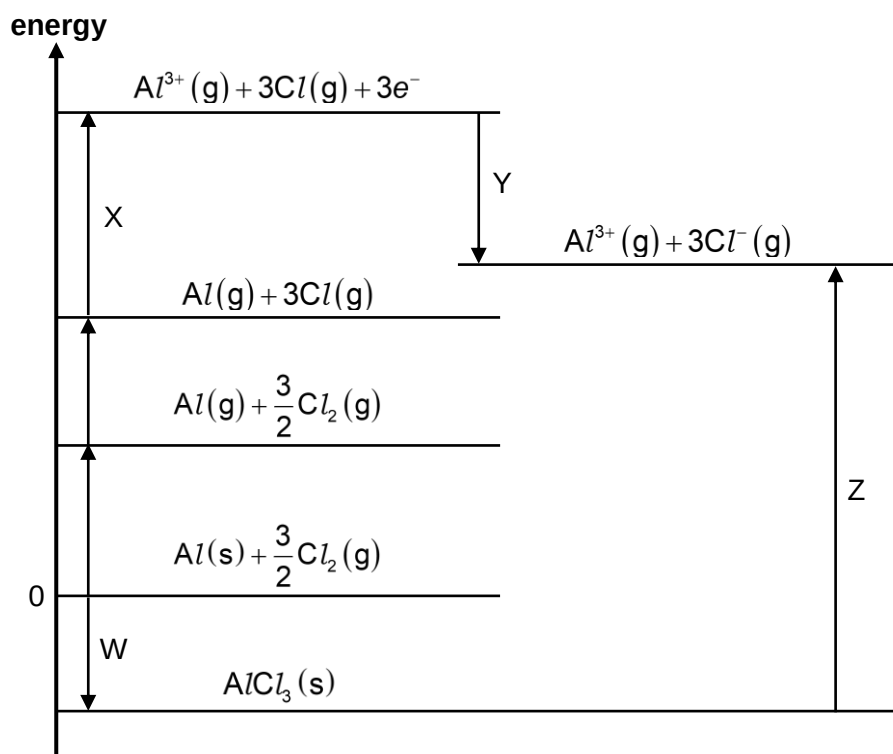


Using the information above, what is the heat liberated for the following neutralisation?



- A 57 kJ B 71 kJ C 114 kJ D 228 kJ

- 12 The Born-Haber cycle for the formation of aluminium chloride is shown below:



Which of the following is correct for this cycle?

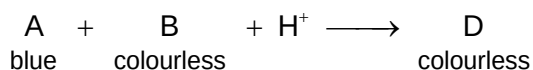
- 1 W is the bond energy of $\text{AlCl}_3(\text{s})$.
- 2 X is the sum of the first three ionisation energies of aluminium.
- 3 Y is the sum of the first three electron affinities of chlorine.
- 4 Z is lattice energy of aluminium chloride.

- A 2 only B 3 and 4 only C 1, 2 and 3 only D 1, 2 and 4 only

- 13** The following pairs contain 1 mole of each substance at the same temperature unless otherwise stated. In which pair does the first substance have a higher entropy than the second substance?

	I	II
A	C(graphite)	C(diamond)
B	NaCl(s)	NaCl(aq)
C	Ar in 1 dm ³ vessel	Ar in 2 dm ³ vessel
D	H ₂ O(l) at 25 °C	H ₂ O(l) at 50 °C

- 14** Compound A reacts with compound B and acid to give product D as shown in the equation below.



A series of experiments was carried out using different volumes of compound A, B and acid. The following results were obtained.

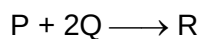
expt	volume of 0.01 mol dm ⁻³ A / cm ³	volume of 0.2 mol dm ⁻³ B / cm ³	volume of 0.02 mol dm ⁻³ H ⁺ / cm ³	volume of water / cm ³	time taken for blue colour to fade / s
1	20	20	20	20	100.0
2	10	20	20	30	100.0
3	20	40	20	0	25.0
4	10	30	30	10	44.4

Which of the following can be concluded from these results?

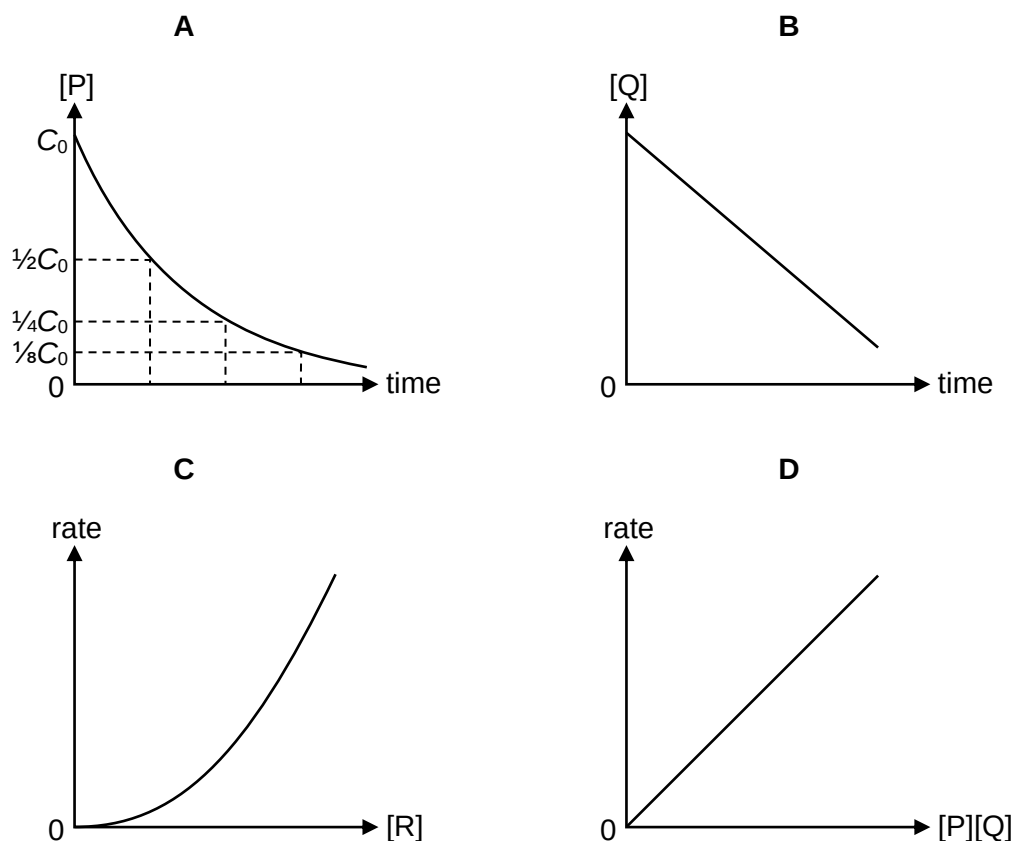
- The overall order of reaction is 3.
- Order of reaction with respect to A is zero.
- When [B] is doubled, $t_{1/2}$ decreased by a factor of 4.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1 only

- 15 0.100 mol dm⁻³ of P and 0.100 mol dm⁻³ of Q is reacted together to form R in the reaction shown below.



Given that the reaction is first order with respect to both P and Q, which graph would be a correct representation of the reaction?



- 16 Copper-64 is a radioactive isotope of copper and has applications for molecular radiotherapy. It decays according to a first order reaction.

If 75% of x mol of copper-64 takes 25.4 hours to decay at a certain temperature, how long will it take for $2x$ moles of copper-64 to decay to 50% at the same temperature?

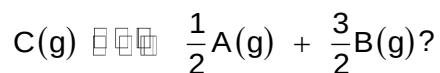
- A** 6.4 h **B** 12.7 h **C** 25.4 h **D** 50.8 h

- 17 Which of the following is not a characteristic of a homogenous catalyst?

- A** It does not affect $\Delta G^\ddagger$ of the reaction.
- B** It speeds up the rate of both the forward and backward reaction to the same extent.
- C** It may be produced at the end of the reaction.
- D** It remains chemically unchanged throughout the reaction.

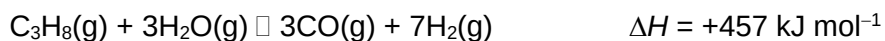
- 18 The reaction $A(g) + 3B(g) \rightleftharpoons 2C(g)$ has a K_c value of 5.

Under the same reaction conditions, what is the value of K_c for the reaction

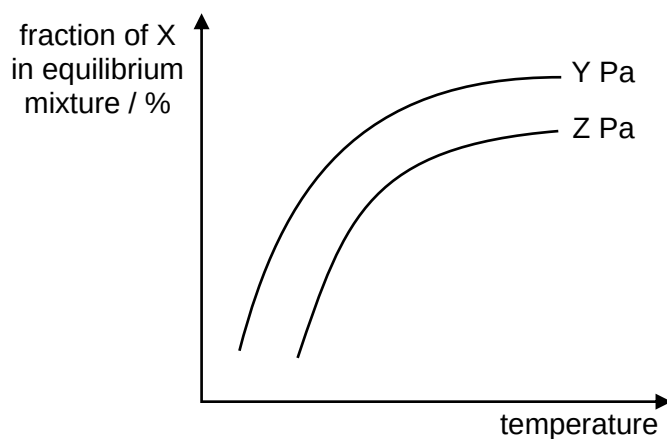


- A $\frac{1}{10}$ B $\frac{2}{5}$ C $\frac{1}{\sqrt{5}}$ D $\frac{\sqrt{5}}{2}$

- 19 X is one of the compounds in the equilibrium mixture.



The graph below shows how the fraction of X varies with temperature at two different pressures.



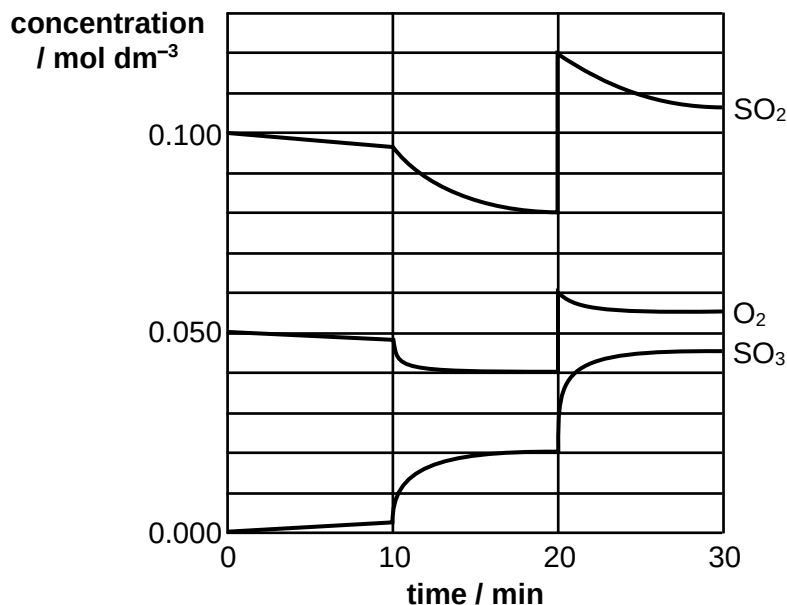
Identify X and the relative magnitudes of Y and Z.

	X	Pressure
A	C_3H_8	$Z > Y$
B	H_2O	$Y > Z$
C	CO	$Z > Y$
D	H_2	$Y > Z$

- 20 During the Contact process, sulfur dioxide is converted to sulfur trioxide as shown by the equation below.

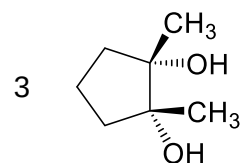
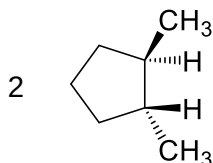
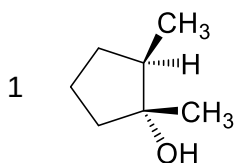


The following graph shows how the concentration of the three gases changed when a series of changes was made.



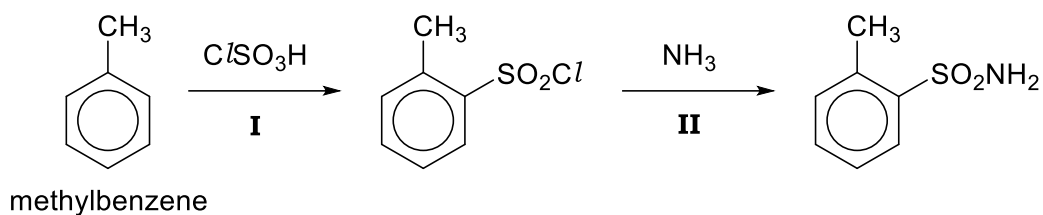
Which of the following describes the change made to the system at the 20th minute?

- A Catalyst was added to the system.
 - B Volume of the reaction vessel decreased.
 - C SO_2 was added to the reaction vessel.
 - D Temperature of the system increased.
- 21 Which of the following molecules are chiral?



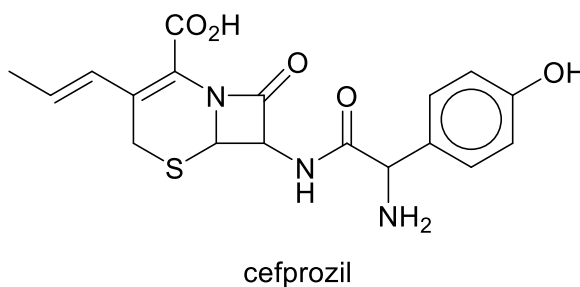
- A 1 and 2 only B 1 and 3 only C 2 and 3 only D 1, 2 and 3

- 22 Consider the following reaction of methylbenzene, $\text{C}_6\text{H}_5\text{CH}_3$.



Which of the following best describes reactions **I** and **II**?

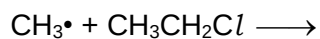
- | | I | II | |
|----------|--------------|--------------|--|
| A | substitution | substitution | |
| B | substitution | addition | |
| C | addition | addition | |
| D | addition | substitution | |
- 23 Cefprozil is a second-generation cephalosporin antibiotic that can be used to treat ear infections, skin infections, and other bacterial infections.



What is the total number of stereoisomers that are possible for this molecule?

- A** 32 **B** 16 **C** 8 **D** 4
- 24 Use of the Data Booklet is relevant to this question.

On the basis of bond energies, what could be the products of the following reaction?



- A** $\text{CH}_4 + \text{CH}_3\text{CHCl}\cdot$
B $\text{CH}_3\text{CH}_2\text{CH}_2\cdot + \text{HCl}$
C $\text{CH}_3\text{CH}_2\text{CH}_3 + \text{Cl}\cdot$
D $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{H}\cdot$

- 25 When 2-methylpropane reacts with limited gaseous bromine under UV light, only two mono-brominated products can be obtained.

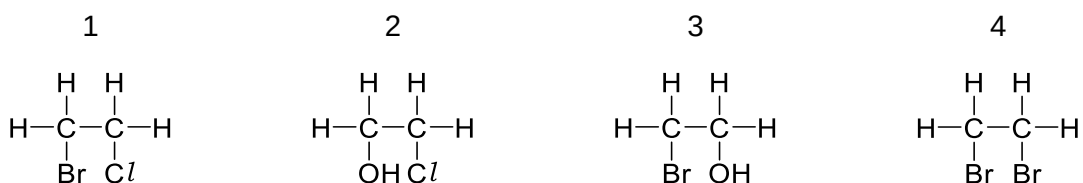
Which of the following statements is false?

- A Both products do not rotate the plane of polarised light.
 B The by-product $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$ can be formed during the reaction.
 C The ratio in which the two mono-brominated products are formed is 1 : 3.
 D The reaction is a radical reaction.
- 26 A catalytic converter is fitted into the exhaust system of a car, so that harmful gases are converted into carbon dioxide, nitrogen and water vapour.

Which reaction is the least likely to take place on the surface of the catalyst?

- A hydrocarbons + oxides of nitrogen $\longrightarrow$ carbon dioxide + water + nitrogen
 B carbon monoxide + oxides of nitrogen $\longrightarrow$ carbon dioxide + nitrogen
 C carbon monoxide + hydrocarbons $\longrightarrow$ carbon dioxide + water
 D nitrogen dioxide $\longrightarrow$ oxygen + nitrogen
- 27 Which statement about the reaction between propene and bromine is true?
- A The mixture of products formed is optically active.
 B The bromine atom attacking the $\text{C}=\text{C}$ is acting as a nucleophile.
 C The hybridisation on the reactive carbon changes from sp^2 to sp^3 .
 D The positively charged carbon in the carbocation intermediate formed contains an unpaired electron.

- 28 The reaction of ethene with bromine in the presence of aqueous sodium chloride gives a mixture of products. Which products will be found in the mixture?



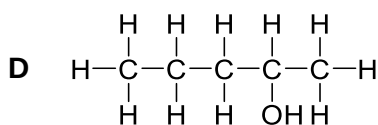
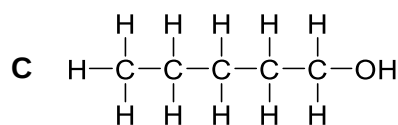
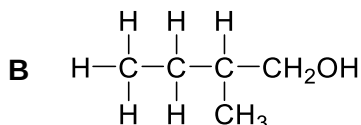
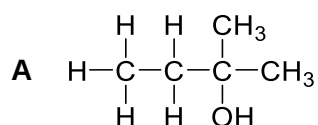
- A 1, 3 and 4 only B 1, 2 and 3 only C 1 and 4 only D 2 and 3 only

- 29 X is an alcohol with the molecular formula $C_5H_{11}OH$.

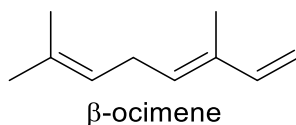
When X undergoes reaction with excess concentrated H_2SO_4 , two isomeric alkenes (ignore *cis-trans* isomers) are formed.

Both alkenes react with hot acidified $KMnO_4$ to form carboxylic acids as the only organic products.

What is the structural formula of X?



- 30 Ocimenes are a group of isomeric hydrocarbons with a sweet herbal scent and are used in perfumes. The structure of one isomer is as shown.



Which statement about β -ocimene is true?

- A** β -ocimene has a total of four stereoisomers.
- B** The major product of the reaction of β -ocimene with HBr has two chiral centres.
- C** In the presence of excess bromine and UV light, β -ocimene undergoes free radical substitution only.
- D** The product of the reaction of β -ocimene with cold dilute alkaline potassium manganate(VII) solution has five chiral centres.